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Optimizing Car Insurance Risk Management In Kenya via Loss Distribution

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for the degree of
Bachelor of Science in Actuarial Science

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Declaration

We hereby declare that this project is our original work and has not been presented before in any other academic institution for any academic award.

Signed:

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With the approval of our supervisor:

Prof. Caroline A. Ogutu ………………………………………

On this……………………………………day of……………………………………. 2024.

Acknowledgement

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Finally, we express our heartfelt gratitude to the Almighty for granting us the resilience and dedication needed to complete this journey.

Abstract

Kenya's car insurance market faces numerous challenges, including high rates of uninsured motorists, fraudulent claims, and inadequate risk assessment practices. This study aims to enhance risk management strategies by analyzing loss distributions and calculating key risk metrics.

By quantifying potential financial risks and extreme loss scenarios, we provide insights for improving pricing models, underwriting criteria, and claims management processes. Applying these findings to real-world data from the Kenyan car insurance industry offers practical recommendations for insurers.

Ultimately, this research contributes to a more resilient and efficient car insurance sector in Kenya, benefiting both insurers and consumers through better risk assessment and management practices.

Contents

Chapter 1

Introduction

Insurance is a financial mechanism that provides protection against potential financial losses by transferring risk from an individual or entity to an insurance company. This risk transfer is facilitated through the payment of premiums, which are pooled together to cover the losses of the insured. Insurance is essential for promoting economic stability and resilience, as it mitigates the financial impact of unforeseen events on individuals and businesses.

The car insurance industry, a critical segment of the broader insurance market, offers financial protection to vehicle owners against losses arising from accidents, theft, and other related incidents. By ensuring that motorists can recover financially from these events, car insurance supports the uninterrupted flow of goods and services, thereby contributing to the overall economic stability and growth of a nation.

In Kenya, the car insurance industry is a vital pillar of the country's economy. It not only provides financial security to motorists but also fosters economic stability and growth by facilitating commerce and mobility. However, despite its importance, the sector faces significant challenges that impede its progress and threaten its sustainability.

1.1 Problem Statement

The Kenyan car insurance market is increasingly plagued by rising incidences of uninsured motorists, escalating fraudulent claims, and inefficiencies in risk management. These issues not only undermine the profitability and sustainability of insurance companies but also jeopardize the financial security of policyholders.

Uninsured motorists pose a significant risk as they contribute to higher premiums for insured drivers and create gaps in financial recovery following accidents. Fraudulent claims inflate costs for insurers, leading to increased premiums and financial strain on the industry. Additionally, inefficiencies in risk management practices result in suboptimal allocation of resources and increased vulnerability to financial losses. There is an urgent need for a thorough analysis of loss distributions to uncover critical insights that can significantly enhance risk management practices. By understanding the frequency and severity of losses, insurers can develop more effective strategies to mitigate risks and improve their financial stability.

1.2 Objectives of the Project

1.2.1 Primary objective

Anlayze loss distributions to optimize car insurance risk management in Kenya.

1.2.2 Specific Objectives

The primary objective is further outlined into the following specific objectives:

1. **Simulate Loss Ratios Using Gamma and Poisson Distributions:**
   - Use the best-fitting Gamma distribution to model the severity of loss events, helping to understand the potential magnitude of individual claims in Kenya's car insurance sector.
   - Use the Poisson distribution to model the frequency of loss events, providing insights into how often claims are likely to occur.
2. **Calculate and Analyze Risk Metrics (VaR and TVaR):**
   - Calculate the Value at Risk (VaR) at a 95% confidence level to quantify the maximum expected loss over a specified time frame.
   - Determine the Tail Value at Risk (TVaR) to measure the expected loss that exceeds the VaR threshold, providing insight into extreme risk scenarios.
3. **Apply Methods to Kenyan Car Insurance Data:**
   - Implement the simulation and risk calculation methods on real-world data from the Kenyan car insurance industry to validate the models and obtain practical insights.

1.3 Significance of The Study

Analyzing loss distributions is crucial for effective risk management in Kenya's car insurance market. This study aims to provide insights that can help insurers refine their pricing models, enhance underwriting criteria, and optimize claims management processes. By fitting suitable loss distributions to Kenyan motor insurance data, the research addresses key challenges such as high rates of uninsured motorists, fraudulent claims, and inadequate risk assessment practices. The findings will contribute to the development of more accurate and reliable risk assessment tools, thereby improving the financial stability and efficiency of Kenya's car insurance sector. This, in turn, benefits both insurers and consumers, promoting economic stability and growth. Additionally, this study adds to the existing body of knowledge in the field, supporting better regulatory practices and risk management strategies in the broader context of the African insurance market.

Chapter 2

Literature Review

The car insurance industry presents a dynamic landscape characterized by diverse risks and challenges that necessitate effective risk management strategies. Central to this endeavor is the analysis of loss distributions, which provides crucial insights into the frequency and severity of losses incurred by insurers. This literature review critically examines existing research on loss distributions in the context of car insurance risk management, focusing on theoretical frameworks, empirical studies, and practical applications relevant to the Kenyan market.

This literature review serves as a foundation for the current research project, which aims to analyze loss distributions and optimize car insurance risk management in Kenya. By synthesizing existing knowledge and identifying gaps in the literature, this review provides valuable insights that guide the research methodology, inform data analysis techniques, and facilitate the development of actionable recommendations for insurers and policymakers.

Loss distributions represent the statistical depiction of the frequency and severity of losses incurred by an entity over a specific period. In the context of car insurance, loss distributions play a crucial role in modeling and analyzing the financial risks associated with insurance claims. Achieng (2010) and Mazviona and Chiduza (2013) conducted seminal studies on loss distributions in the African insurance market, providing valuable insights into modeling claim severity data using parametric distributions such as the lognormal and Pareto distributions.

The frequency distribution component of loss distributions quantifies the likelihood of experiencing different numbers of loss events within a specified time period. In the context of car insurance, the frequency distribution provides insights into the probability of various types of accidents occurring per year, thereby enabling insurers to assess their exposure to different risk scenarios. The severity distribution component characterizes the financial magnitude of individual loss events, representing the distribution of claim amounts or losses incurred per event. Achieng (2010) and Mazviona and Chiduza (2013) employed maximum likelihood estimation and goodness-of-fit tests to model claim severity data using parametric distributions such as the lognormal and Pareto distributions, highlighting their suitability for different types of claims.

Building on the theoretical and empirical frameworks established in past studies, the current research project seeks to extend the analysis of loss distributions to the Kenyan economy. By leveraging company-specific data sets and employing advanced modeling techniques such as Monte Carlo simulation, the study aims to fit suitable loss distributions to claim severity data for motor insurance companies in Kenya. In addition to empirical studies, theoretical frameworks underpin the analysis of loss distributions and risk management in the insurance industry. The economic theory of insurance, for instance, emphasizes the role of risk pooling and diversification in mitigating financial losses for policyholders. By spreading risks across a large pool of insured individuals, insurers can stabilize their financial performance and ensure the availability of funds to cover claims.

Similarly, behavioral economics offers insights into the psychological factors influencing insurance decisions and risk perceptions. Prospect theory, for example, suggests that individuals tend to exhibit loss aversion, placing greater weight on potential losses than equivalent gains. Understanding these behavioral biases is essential for designing effective risk management strategies and promoting informed decision-making among policyholders.

Numerous empirical studies have investigated loss distributions and risk management practices in various insurance markets worldwide. While some studies focus on specific types of insurance, such as motor insurance or property insurance, others adopt a broader perspective to analyze aggregate loss distributions across multiple lines of business.

Research by Cummins and Nini (2002) examined the distribution of property-liability insurance losses in the United States, highlighting the significance of tail risk and the potential impact of catastrophic events on insurer solvency. Similarly, Frees et al. (2015) conducted a comprehensive analysis of healthcare insurance claims, demonstrating the importance of predictive modeling techniques in assessing healthcare risk and pricing insurance premiums.

Despite the wealth of research on loss distributions and insurance risk management, there is a dearth of literature specifically addressing these issues in the Kenyan context. Kenya's car insurance market presents unique challenges, including high rates of uninsured motorists, fraudulent claims, and inadequate risk assessment practices. By applying insights from global research to the Kenyan economy, policymakers and insurers can develop tailored strategies to address these challenges and improve the overall stability and efficiency of the insurance market.

The findings of empirical studies and theoretical frameworks have significant implications for insurers, policymakers, and consumers in Kenya. By better understanding the frequency and severity of potential losses, insurers can enhance their pricing models, underwriting criteria, and claims management processes to improve profitability and solvency. Policymakers can use insights from empirical research to enact regulations that promote transparency, competition, and consumer protection within the insurance industry. Additionally, consumers stand to benefit from more affordable premiums and better coverage options resulting from improved risk management practices.

While existing research provides valuable insights into loss distributions and risk management practices, there are several avenues for future research in the Kenyan context. Longitudinal studies tracking changes in loss distributions over time could provide valuable insights into emerging trends and risk factors affecting the insurance market. Additionally, research focusing on innovative risk transfer mechanisms, such as parametric insurance products or microinsurance schemes, could help expand insurance coverage and promote financial inclusion among underserved populations in Kenya.

Chapter 3

Methodology

3.1 Data Collection and Preparation

The initial phase of the project involves gathering data relevant to the Kenyan car insurance market.

To ensure the data is ready for analysis, the following steps will be taken:

• Loading the Data: Import the dataset into R for analysis. This involves reading the CSV file and converting it into a dataframe format suitable for manipulation and analysis in R.

• Handling Missing Values: Identify and impute missing values. For numeric columns, the mean of the column will be used for imputation. This step is essential to maintain the integrity of the dataset and ensure that the absence of data does not skew the analysis.

• Data Type Conversion: Ensure all numeric data is properly formatted as numeric types. This is necessary for accurate calculations and statistical analysis.

3.2 Exploratory Data Analysis(EDA)

EDA is crucial to understand the structure and characteristics of the data. This involves:

• Descriptive Statistics: Calculate summary statistics (mean, median, standard deviation, etc.) for all columns to understand their distributions and central tendencies. This provides a foundational understanding of the dataset and highlights any anomalies.

• Visualization: Create histograms for each numeric column to visualize their distributions. This includes Earned Premium Income, Exposure (Years), Total Paid, Estimated Cost, Cost, Number of Claims, and Number of Large Claims. Visualizations help in identifying the shape of the data distribution and spotting any outliers or skewness.

Mean

The mean (or average) of a dataset is calculated by summing all the data points and dividing by the number of data points. If we have a dataset $X = \{x_1, x_2, \ldots, x_n\}$, the mean μ is given by:

$$\mu = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{3.1}$$

where:

- n is the number of data points.
- x_i represents each individual data point.

Median

The median is the middle value of the dataset when it is ordered in ascending or descending order. For a dataset sorted in ascending order, the median M is given by:

- If n is odd, $M = x_{\frac{n+1}{2}}$.
- If n is even, $M = \frac{1}{2}(x_{\frac{n}{2}} + x_{\frac{n}{2}+1})$.

Standard Deviation

The standard deviation measures the amount of variation or dispersion in a dataset. It is calculated as the square root of the variance. The variance σ^2 is given by:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2 \tag{3.2}$$

The standard deviation σ is then:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2} \tag{3.3}$$

where:

- μ is the mean of the dataset.
- x_i represents each individual data point.
- n is the number of data points.
- σ^2 is the variance of the dataset.
- σ is the standard deviation of the dataset.

3.3 Calculation of Key Metrics

To gain insights into the performance and risk of the insurance portfolio, key metrics will be calculated:

Loss Ratio

The Loss Ratio is a measure of the proportion of earned premiums that are paid out as claims. It is an indicator of the profitability of an insurance portfolio. The formula for the Loss Ratio LR is:

$$LR = \frac{\text{Total Paid}}{\text{Earned Premium Income}} \tag{3.4}$$

where:

- **Total Paid** is the total amount paid out in claims.
- **Earned Premium Income** is the total premium income earned during the period.

A lower Loss Ratio indicates higher profitability, as a smaller proportion of premiums is being paid out as claims.

Frequency

Frequency measures how often claims occur relative to the exposure period. It provides an understanding of the rate at which claims are made. The formula for Frequency F is:

$$F = \frac{\text{Number of Claims}}{\text{Exposure (Years)}} \tag{3.5}$$

where:

- **Number of Claims** is the total number of claims filed.
- **Exposure (Years)** represents the total exposure period, typically measured in years.

A higher Frequency indicates a higher occurrence of claims relative to the exposure period.

Severity

Severity indicates the average cost per claim, providing insight into the financial impact of each claim. The formula for Severity S is:

$$S = \frac{\text{Total Paid}}{\text{Number of Claims}} \tag{3.6}$$

where:

- **Total Paid** is the total amount paid out in claims.
- **Number of Claims** is the total number of claims filed.

3.3.1 Expected Results and Importance

Loss Ratio:
• The Loss Ratio is crucial for assessing the overall profitability and efficiency of the insurer.
• Expected to be below 1 (or 100 percent) for profitability. If it exceeds 1, the insurer is paying out more in claims than it is earning in premiums, indicating potential financial distress.

Frequency:
• Understanding claim frequency helps insurers predict the likelihood of future claims
• Lower frequency indicates fewer claims per exposure year, which is preferable for risk management.
• High frequency might suggest a need for more stringent underwriting policies or higher premiums.

Severity:
• Understanding claim severity helps in assessing the potential financial burden of claims.
• Lower severity is preferable as it indicates lower average claim costs.
• High severity requires careful reserve planning and might suggest a need for better risk mitigation strategies.
A higher Severity indicates a greater financial impact per claim.

3.4 Fitting Probability Distributions for Frequency and Severity of Claims

To model the frequency and severity of losses in the car insurance sector, appropriate probability distributions will be fitted. This step is critical for understanding the underlying risk dynamics and for predicting future outcomes.

Frequency of Claims

Poisson Distribution

The Poisson distribution is commonly used for modeling count data, particularly when events are rare and occur independently. The probability mass function (PMF) of a Poisson distribution is given by:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots \tag{3.7}$$

where:

- λ is the average rate (mean) of occurrences in a fixed interval.
- k is the number of occurrences.

In the context of car insurance, λ represents the average number of claims per unit of exposure (e.g., per year).

Severity of Claims

Gamma Distribution

The Gamma distribution is suitable for modeling positive continuous data with right skew, which is typical for claim severity. The probability density function (PDF) of a Gamma distribution is given by:

$$f(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0 \tag{3.8}$$

where:

- α is the shape parameter.
- β is the rate parameter (inverse scale parameter).
- $\Gamma(\alpha)$ is the Gamma function.

In car insurance, α and β describe the distribution of claim amounts, helping to understand the variability and typical size of claims.

3.5 Goodness-of-Fit Testing

To evaluate how well the chosen distributions fit the data, goodness-of-fit tests will be conducted. One commonly used test is the Chi-Square Test.

3.5.1 Chi-Square Test

The Chi-Square Test is used to compare the observed and expected frequencies under the fitted distributions. This helps in assessing the adequacy of the fitted model.

Test Statistic

The test statistic for the Chi-Square Test is calculated as:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \tag{3.9}$$

where:

- O_i is the observed frequency for the i-th category.
- E_i is the expected frequency for the i-th category, calculated under the fitted distribution.
- k is the number of categories.

Degrees of Freedom

The degrees of freedom for the Chi-Square Test is given by:

$$\mathrm{df} = k - p - 1 \tag{3.10}$$

where:

- k is the number of categories.
- p is the number of parameters estimated from the data.

Interpreting the Chi-Square Test in Car Insurance

In the context of car insurance, the Chi-Square Test can be used to assess how well the fitted probability distributions (e.g., Poisson, Negative Binomial, Gamma, Lognormal) match the observed data for claim frequency and severity.

Steps to perform the Chi-Square Test:

1. Categorize the Data: Divide the data into k categories (e.g., ranges of claim amounts or intervals of claim counts).

2. Calculate Observed Frequencies: Determine the observed frequency O_i for each category.

3. Calculate Expected Frequencies: Compute the expected frequency E_i for each category using the fitted distribution.

4. Compute the Test Statistic: Use the formula to calculate χ^2.

5. Compare with Critical Value: Compare the calculated χ^2 value with the critical value from the Chi-Square distribution table with df degrees of freedom.

If the calculated χ^2 value is less than the critical value, we fail to reject the null hypothesis, indicating that the fitted distribution is adequate. Otherwise, we reject the null hypothesis, suggesting that the fitted distribution may not be a good fit for the data.

3.6 Risk Analysis

Using the best-fitting distributions, we can conduct a risk analysis to assess the potential financial risk associated with the insurance portfolio.

3.6.1 Simulation of Loss Ratios

To simulate loss ratios based on the best-fitting distribution (e.g., Gamma), we generate random samples from the fitted distribution to estimate the range and likelihood of different loss ratios.

Gamma Distribution for Loss Ratios

Assume that the loss ratios follow a Gamma distribution with shape parameter α and rate parameter β. The probability density function (PDF) of the Gamma distribution is:

$$f(x;\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0 \tag{3.11}$$

To simulate loss ratios:

- Generate a large number of random samples $x_1, x_2, \ldots, x_n$ from the Gamma distribution.
- Use these samples to estimate the range and likelihood of different loss ratios.

3.6.2 Value at Risk (VaR)

Definition and Interpretation

Value-at-Risk (VaR) is a fundamental risk measure used by financial institutions, insurance companies, and other enterprises to quantify the potential for significant losses over a given period. It answers the question: "What is the probability of an adverse outcome?"

- **Mathematical Definition:** VaR at the 100pth security level, denoted by $\mathrm{VaR}_p(X)$, is the 100pth percentile of the loss distribution X. Formally, it is the value π_p such that:

  $$P(X > \pi_p) = 1 - p \tag{3.12}$$

  This implies that there is a $1 - p$ probability that the loss X will exceed π_p.

- **Interpretation in Risk Management:** VaR represents the threshold value of losses that will not be exceeded with a specified high level of confidence (e.g., 95

Usage and Security Levels

- **Enterprise-Level Security:** Higher security levels (e.g., 99
- **Sub-Unit Security:** Lower security levels (e.g., 95

Advantages of VaR

- **Clear and Simple Definition:** VaR has a straightforward definition as a quantile of the loss distribution, making it intuitive and easy to understand.
- **Closed-Form Evaluation:** For certain probability distributions, VaR can be calculated analytically. This is particularly useful when dealing with standard loss models where the percentiles can be derived directly.
- **Software Evaluation:** For more complex loss distributions that do not have a closed form for percentiles, VaR can be evaluated using statistical software and numerical methods, making it versatile for various applications.

Limitations of VaR

- **Tail Risk Ignorance:** VaR does not provide information about the magnitude of losses beyond the VaR threshold. This limitation means that it can underestimate the risk of extreme losses.
- **Non-Subadditivity:** VaR is not necessarily subadditive, meaning that the VaR of a combined portfolio might exceed the sum of the VaRs of individual portfolios. This property complicates risk aggregation across different business units or asset classes.
- **Focus on a Single Quantile:** VaR focuses on a specific quantile of the loss distribution, potentially ignoring other important aspects of the distribution, such as skewness or kurtosis.
- **Assumption of Continuous Distributions:** The straightforward application of VaR assumes continuous loss distributions, which may not always be the case in practical scenarios with discrete or mixed distributions.

3.6.3 Tail Value at Risk (TVaR)

Tail-Value-at-Risk (TVaR) is a risk measure that provides more information about the tail of the loss distribution than VaR. While VaR is a cutoff point, TVaR reflects the shape of the tail beyond the VaR threshold.

Definition-based Formulation

$$\mathrm{TVaR}_p(X) = \mathbb{E}(X \mid X > \pi_p) = \frac{\int_{\pi_p}^{\infty} x f(x)\, dx}{1 - F(\pi_p)} = \frac{\int_{\pi_p}^{\infty} x f(x)\, dx}{1 - p} \tag{3.13}$$

This formula defines TVaR as the expected loss given that the loss exceeds the VaR threshold π_p.

Average of VaRs Formulation

$$\mathrm{TVaR}_p(X) = \frac{\int_{p}^{1} \mathrm{VaR}_w(X)\, dw}{1 - p} \tag{3.14}$$

This formulation illustrates TVaR as the average of all VaR values above the threshold p. It provides insight into the tail behavior beyond the specific VaR level by averaging the risks over a range of probabilities from p to 1.

Mean Excess Loss Formulation

$$\mathrm{TVaR}_p(X) = \pi_p + \frac{\int_{\pi_p}^{\infty} (x - \pi_p) f(x)\, dx}{1 - p} = \mathrm{VaR}_p(X) + e(\mathrm{VaR}_p(X)) \tag{3.15}$$

Here, $e(x)$ represents the mean excess loss function evaluated at x. This shows that TVaR is the sum of VaR and the average excess loss above the VaR threshold. It indicates that TVaR accounts for both the threshold value (VaR) and the tail distribution's shape beyond VaR.

Parametric Formulation

$$\mathrm{TVaR}_p(X) = \mathrm{VaR}_p(X) + e(\mathrm{VaR}_p(X)) = \mathrm{VaR}_p(X) + \frac{\mathbb{E}[X] - \mathbb{E}[X \wedge \mathrm{VaR}_p(X)]}{1-p} \tag{3.16}$$

This formulation expresses TVaR in terms of the expected value $\mathbb{E}[X]$ and the expected truncated loss $\mathbb{E}[X \wedge \mathrm{VaR}_p(X)]$. It highlights how TVaR incorporates the overall expectation of losses and adjusts it based on the losses up to the VaR level.

Implications of TVaR in Risk Management

- **Comprehensive Risk Measure:** Unlike VaR, which only provides a cutoff point for risk, TVaR offers a fuller picture by considering the tail behavior beyond the VaR threshold. This makes TVaR a more robust measure for understanding extreme losses.
- **Better Insight into Tail Risk:** By incorporating the mean excess loss, TVaR captures the severity of losses that exceed the VaR threshold, providing crucial information for risk assessment and management.
- **Useful for Extreme Event Modeling:** TVaR is particularly valuable in fields where understanding extreme events is critical, such as finance, insurance, and environmental studies.

By utilizing these formulations, risk managers can gain a deeper understanding of potential extreme losses, leading to more effective risk mitigation strategies and more informed decision-making.

Chapter 4

DATA ANALYSIS

4.1 Data collection

The dataset, named $comprehensive_set.csv$, includes various attributes such as Earned Premium Income, Exposure (Years), Total Paid, Estimated Cost, Cost, Number of Claims, and Number of Large Claims.
This dataset is crucial as it provides a comprehensive view of the insurance portfolio, capturing both the financial aspects (earned premiums, costs) and the operational aspects (exposure, claims).

The data was sourced from NextCar Insurance - Car Insurance Kenya Statistics. This source provides detailed and reliable statistics that are essential for our analysis.
Here is the link to the data source:

`https://data.world/nextcarinsurance/car-insurance-kenya-statistics`

4.2 Exploratory Data Analysis

Table 1: Descriptive Statistics

Statistic	Earned Premium Income ('000)	Exposure (years)	Total Paid ('000)	Estimated Cost ('000)	Cost ('000)	No. of Claims	No. of Large Claims
Length	24	24	24	24	24	24	24
Minimum	397	616	95	117.0	95	32.0	0.00
1st Quartile	4615	4190	2498	583.8	3444	338.8	4.50
Median	8286	11240	5794	1783.5	7769	1050.5	12.00
Mean	29531	57964	15681	3054.5	18481	3390.2	26.54
3rd Quartile	32929	55991	18231	3680.8	20645	3241.5	33.25
Maximum	147584	330355	82274	16994.0	94802	19742.0	135.00
NA's			2				

Figure 4.1: descriptive statistics

Histograms were utilized to visualize the distribution of key numerical variables in the dataset. This helped us gain insights into the data's underlying structure and identify any patterns or anomalies.

4.2.1 Earned Premium Income

The histogram for Earned Premium Income provides a visual representation of the distribution of premium income earned by the insurance company.
The x-axis represents the premium income values, while the y-axis represents the frequency of occurrences for these values

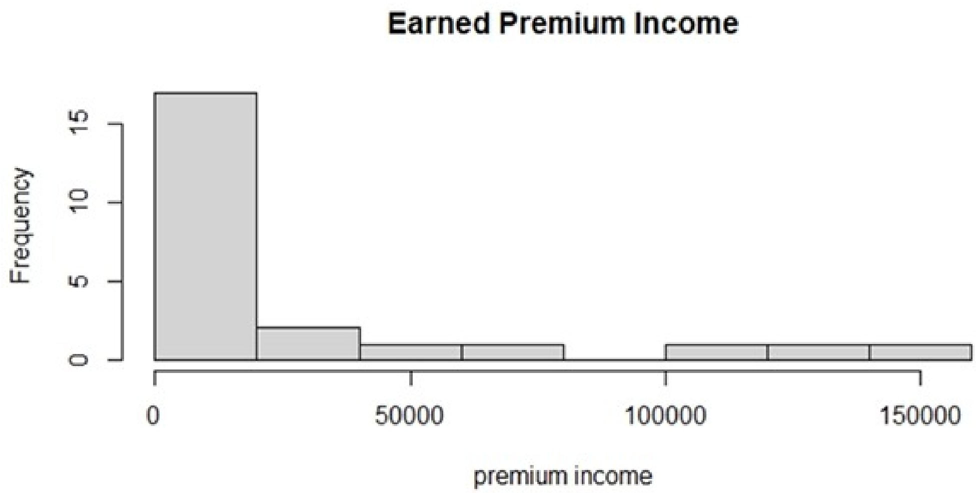

Figure 4.2: earned premium income

Insights:

1. **Right-Skewed Distribution:**
   - The histogram is highly right-skewed, indicating that most of the premium income values are concentrated towards the lower end of the scale.
   - This skewness suggests that a significant portion of the policies have relatively lower premium incomes.
2. **High Frequency of Low Values:**
   - The majority of the premium income values fall within the lower range, as seen by the tall bar on the left side of the histogram.
   - This implies that many policies are generating smaller amounts of premium income.
3. **Presence of High-Value Policies:**
   - There are a few bars towards the right, representing higher premium income values.
   - These bars indicate the presence of some high-value policies, although they are less frequent compared to low-value policies.

Implications for Analysis:

- The concentration of low premium income values might influence the overall risk assessment and pricing strategies.

- The presence of high-value policies, although infrequent, can have a significant impact on the company's financial performance and risk profile.

4.2.2 Exposure (Years)

The histogram for Exposure (Years) represents the distribution of exposure periods in the dataset. The x-axis shows the duration of exposure in years, while the y-axis indicates the frequency of occurrences for these values.

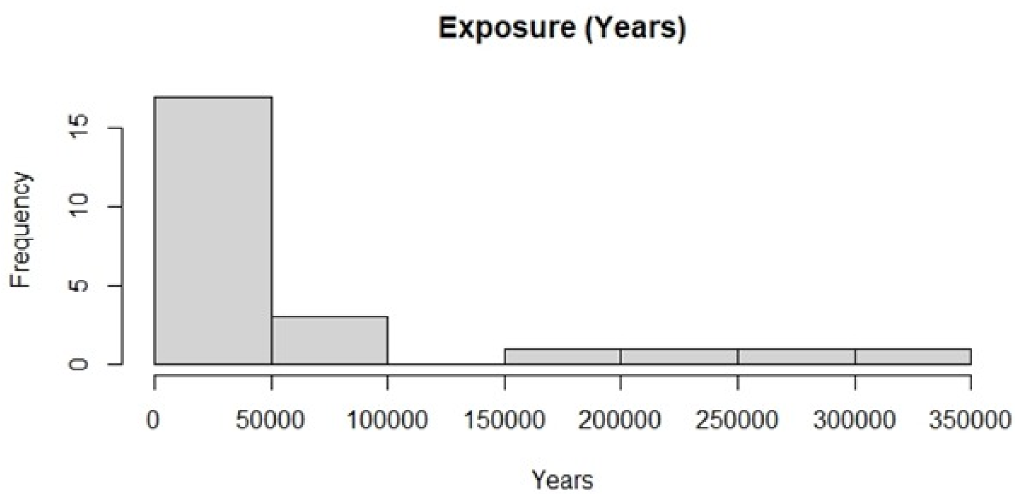

Figure 4.3: Exposure in years

Insights:

1. **Right-Skewed Distribution:**
    - Similar to the Earned Premium Income histogram, the Exposure histogram is also right-skewed, with the majority of exposure periods concentrated towards the lower end of the scale.
    - This indicates that most policies have shorter exposure periods.
2. **High Frequency of Short Exposure Periods:**
    - The tallest bar on the left side of the histogram represents the highest frequency of short exposure periods, suggesting that many policies have relatively short durations.
    - This could imply a higher turnover rate of policies or a focus on short-term coverage.

3. **Presence of Long Exposure Periods:**
   - There are a few bars towards the right, representing longer exposure periods, although these are less frequent.
   - These longer durations might correspond to more stable or long-term policies, which could have different risk profiles compared to short-term policies.

Implications for Analysis:

- The prevalence of short exposure periods might affect the overall stability and predictability of the risk assessment models.
- Understanding the distribution of exposure periods is crucial for pricing strategies, as policies with different durations might have different risk characteristics.
- The presence of both short and long exposure periods suggests a diverse portfolio, which could influence the design and marketing of insurance products.

4.2.3 Total Paid

The histogram for Total Paid represents the distribution of the total amount paid for claims in the dataset. The x-axis shows the total amount paid (in thousands of currency units), while the y-axis indicates the frequency of occurrences for these amounts.

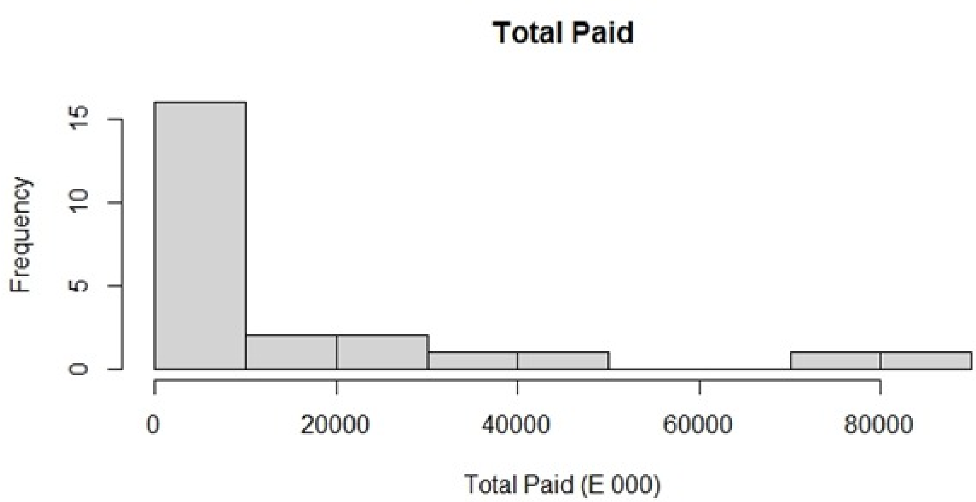

Figure 4.4: Total Paid

Insights:

1. **Right-Skewed Distribution:**
   - The histogram is heavily right-skewed, indicating that the majority of total paid amounts are concentrated towards the lower end of the scale.
   - Most claims result in lower payouts, with a few instances of very high payouts.
2. **High Frequency of Low Payouts:**
   - The tallest bar on the left side of the histogram represents the highest frequency of low payout amounts.
   - This suggests that many claims are of smaller amounts, which could be indicative of frequent but minor claims.
3. **Presence of High Payouts:**
   - There are a few bars towards the right, representing high payout amounts, although these are less frequent.
   - These higher payouts might correspond to severe claims or large losses, which have significant financial impacts on the insurer.

Implications for Analysis:

- The prevalence of low payout amounts might affect the overall financial stability and loss ratios of the insurer.
- Understanding the distribution of claim payouts is crucial for risk assessment, pricing strategies, and reserve allocation.
- The presence of both low and high payouts suggests a need for different risk management strategies for minor and severe claims.

4.2.4 Estimated Cost

The histogram for Estimated Cost represents the distribution of the estimated costs of claims in the dataset. The x-axis shows the estimated cost (in thousands of currency units), while the y-axis indicates the frequency of occurrences for these estimated costs.

Insights:

- **Right-Skewed Distribution:**
  - The histogram is heavily right-skewed, indicating that the majority of estimated costs are concentrated towards the lower end of the scale.
  - Most claims result in lower estimated costs, with a few instances of very high estimated costs.

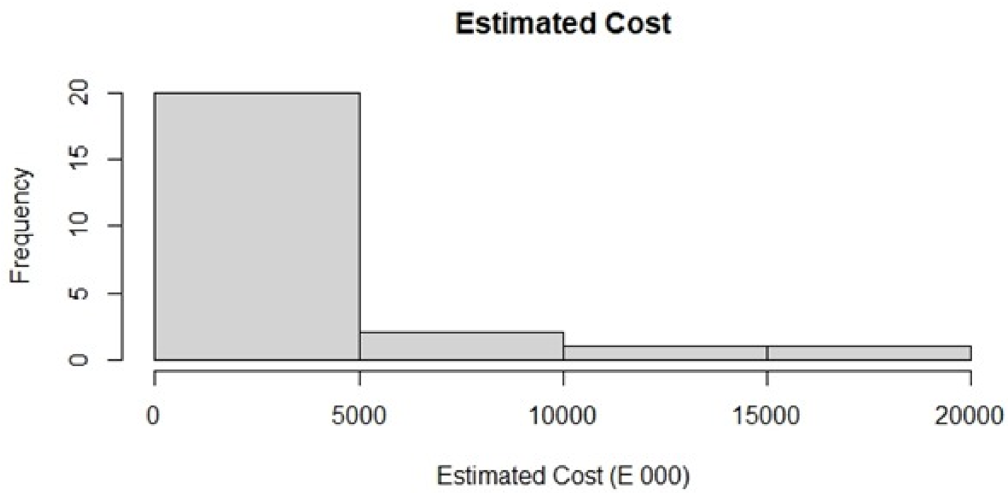

Figure 4.5: Estimated Cost

- **High Frequency of Low Estimated Costs:**
  - The tallest bar on the left side of the histogram represents the highest frequency of low estimated costs.
  - This suggests that many claims have smaller estimated costs, which could be indicative of frequent but minor damages.
- **Presence of High Estimated Costs:**
  - There are a few bars towards the right, representing high estimated costs, although these are less frequent.
  - These higher estimated costs might correspond to severe claims or major damages, which have significant financial impacts on the insurer.

Implications for Analysis:

- The prevalence of low estimated costs might affect the overall financial stability and loss ratios of the insurer. Efficient processing of these frequent low-cost claims can help maintain cost-effectiveness.
- Understanding the distribution of estimated costs is crucial for risk assessment, pricing strategies, and reserve allocation. By identifying patterns in the estimated costs, insurers can better predict future liabilities and adjust premiums accordingly.

- The presence of both low and high estimated costs suggests a need for different risk management strategies for minor and severe claims. For example, minor claims might benefit from streamlined processing, while severe claims could require more thorough investigations and higher reserves to cover potential payouts.

4.2.5 Cost

The histogram for Cost represents the distribution of the actual costs of claims in the dataset. The x-axis shows the cost (in thousands of currency units), while the y-axis indicates the frequency of occurrences for these costs.

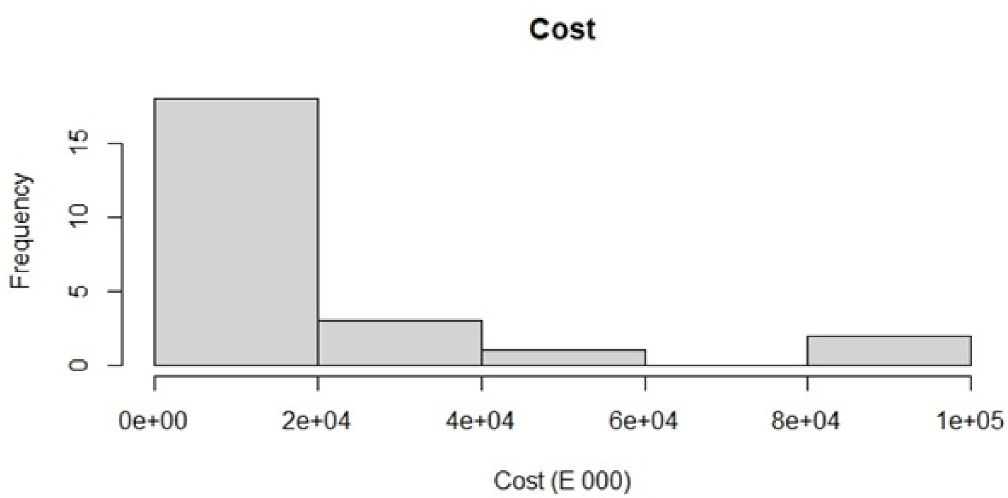

Figure 4.6: Cost

Insights:

- **Right-Skewed Distribution:**
  - The histogram is heavily right-skewed, indicating that the majority of actual costs are concentrated towards the lower end of the scale.
  - Most claims result in lower costs, with a few instances of very high costs.
- **High Frequency of Low Costs:**
  - The tallest bar on the left side of the histogram represents the highest frequency of low cost amounts.

  - This suggests that many claims have smaller costs, which could be indicative of frequent but minor damages or losses.

- **Presence of High Costs:**
  - There are a few bars towards the right, representing high cost amounts, although these are less frequent.
  - These higher costs might correspond to severe claims or major damages, which have significant financial impacts on the insurer.

Implications for Analysis:

- The prevalence of low cost amounts might affect the overall financial stability and loss ratios of the insurer. Efficient processing of these frequent low-cost claims can help maintain cost-effectiveness.
- Understanding the distribution of claim costs is crucial for risk assessment, pricing strategies, and reserve allocation. By identifying patterns in the actual costs, insurers can better predict future liabilities and adjust premiums accordingly.
- The presence of both low and high costs suggests a need for different risk management strategies for minor and severe claims. For example, minor claims might benefit from streamlined processing, while severe claims could require more thorough investigations and higher reserves to cover potential payouts.

4.2.6 Number of Claims

The histogram for Number of Claims represents the distribution of the number of claims filed by policyholders. The x-axis shows the number of claims, while the y-axis indicates the frequency of policyholders who filed these claims

Insights:

- **Right-Skewed Distribution:**
  - The histogram is heavily right-skewed, indicating that the majority of policyholders file a low number of claims.
  - Most policyholders file fewer claims, with a few filing a large number of claims.
- **High Frequency of Low Number of Claims:**
  - The tallest bar on the left side of the histogram represents the highest frequency of low numbers of claims.
  - This suggests that many policyholders either do not file claims frequently or file a single claim.
- **Presence of Multiple Claims:**

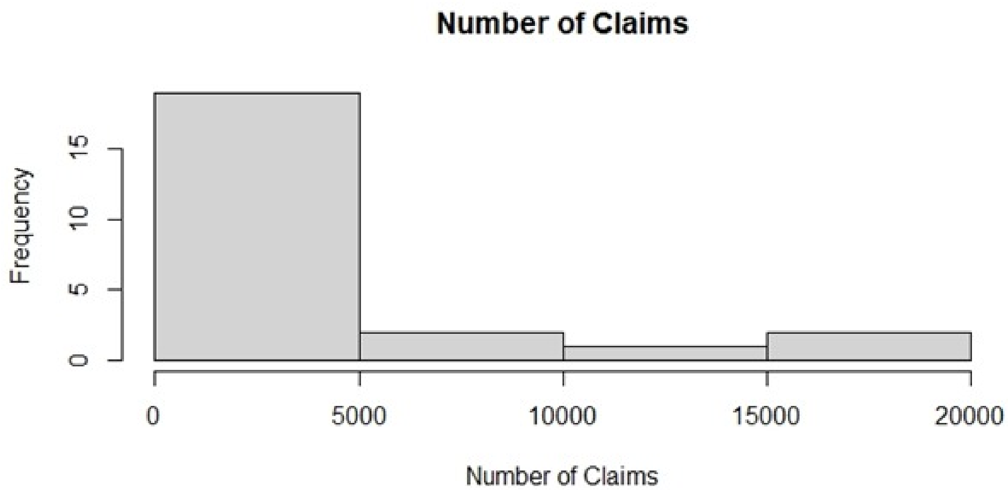

Figure 4.7: Number of Claims

- There are a few bars towards the right, representing policyholders who file a higher number of claims, although these are less frequent.
- These multiple claims could indicate higher risk individuals or those involved in repeated incidents.

Implications for Analysis:

- The prevalence of low numbers of claims suggests that the insurer might have a large number of low-risk policyholders who either drive safely or do not report minor incidents.
- Identifying high-frequency claim filers can help in designing targeted risk mitigation strategies and premium adjustments. Policyholders who file multiple claims might be charged higher premiums or offered specific risk management programs.
- Understanding the distribution of the number of claims aids in resource allocation and customer relationship management. For instance, knowing the pattern of claims can help insurers allocate adequate resources for processing and settling claims efficiently.
- The presence of both low and high numbers of claims suggests a need for differentiated approaches in handling various segments of policyholders. Low-frequency claimants might benefit from standard insurance products, while high-frequency claimants could require tailored risk assessment and management strategies.

4.2.7 Number of Largre Claims

The histogram for the Number of Large Claims represents the distribution of large claims made by policyholders. The x-axis shows the number of large claims, while the y-axis indicates the frequency of occurrences for these claim counts.

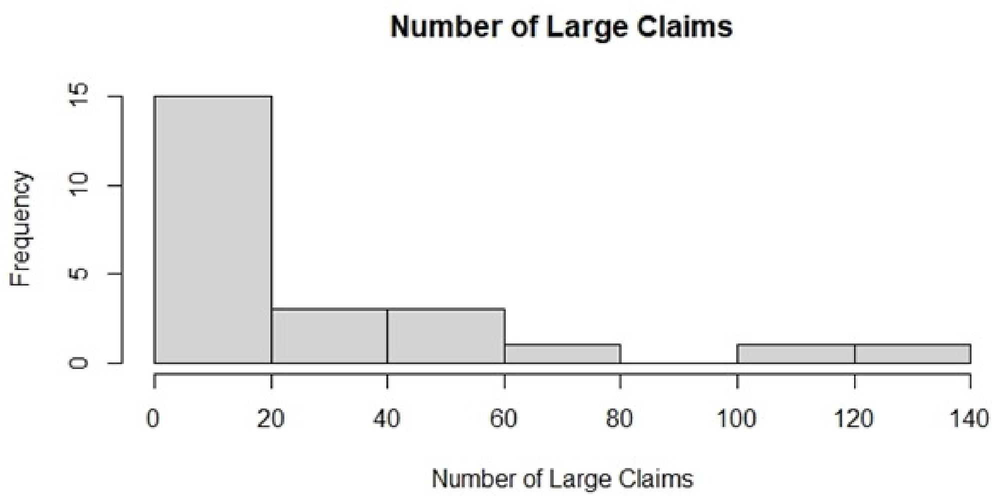

Figure 4.8: Number of Large Claims

Insights:

1. **Right-Skewed Distribution:**
   - **Observation:** The histogram is heavily right-skewed, indicating that the majority of large claim occurrences are concentrated towards the lower end of the scale.
   - **Interpretation:** Most policyholders have a smaller number of large claims, with fewer policyholders having a larger number of large claims.
2. **High Frequency of Few Large Claims:**
   - **Observation:** The tallest bar on the left side of the histogram represents the highest frequency of occurrences with few large claims, particularly around 0-20 large claims.
   - **Interpretation:** This suggests that many policyholders either have no large claims or very few large claims, indicating that large claims are relatively rare events for most policyholders.
3. **Presence of Multiple Large Claims:**

- **Observation:** There are fewer bars as the number of large claims increases, representing policyholders who have multiple large claims, although these are less frequent.
- **Interpretation:** Policyholders with multiple large claims might represent higher risk profiles or more severe incidents, which could require special attention in risk management.

Implications for Analysis:

1. **Risk Concentration:**
   - **Insight:** The prevalence of few large claims among most policyholders highlights the need for effective risk mitigation strategies to handle rare but potentially high-impact events.
   - **Action:** Insurers may need to allocate resources and reserves to cover these infrequent but significant large claims.
2. **Policy Design:**
   - **Insight:** Understanding that most policyholders have few large claims can help in designing policies that incentivize safer behavior and lower claims frequency.
   - **Action:** Offering discounts or benefits for claim-free periods or minimal claims could encourage policyholders to maintain lower risk levels.
3. **Tail Risk Management:**
   - **Insight:** The presence of policyholders with multiple large claims indicates a tail risk that needs careful management.
   - **Action:** Developing specialized products or premium adjustments for high-risk policyholders can help manage the financial impact of these large claims.

4.3 Calculation of Key Metrics

1. Loss Ratio

Formula:

$$\text{Loss Ratio} = \frac{\text{Total Paid}}{\text{Earned Premium Income}} \tag{4.1}$$

Average Value:

- **Loss Ratio:** 0.65

Explanation:

- The Loss Ratio measures the proportion of earned premium income that is paid out in claims.
- A loss ratio of 0.65 indicates that 65% of the earned premiums are used to pay claims.
- This implies that the insurer is retaining 35% of the earned premiums for other expenses, profits, or reserves.

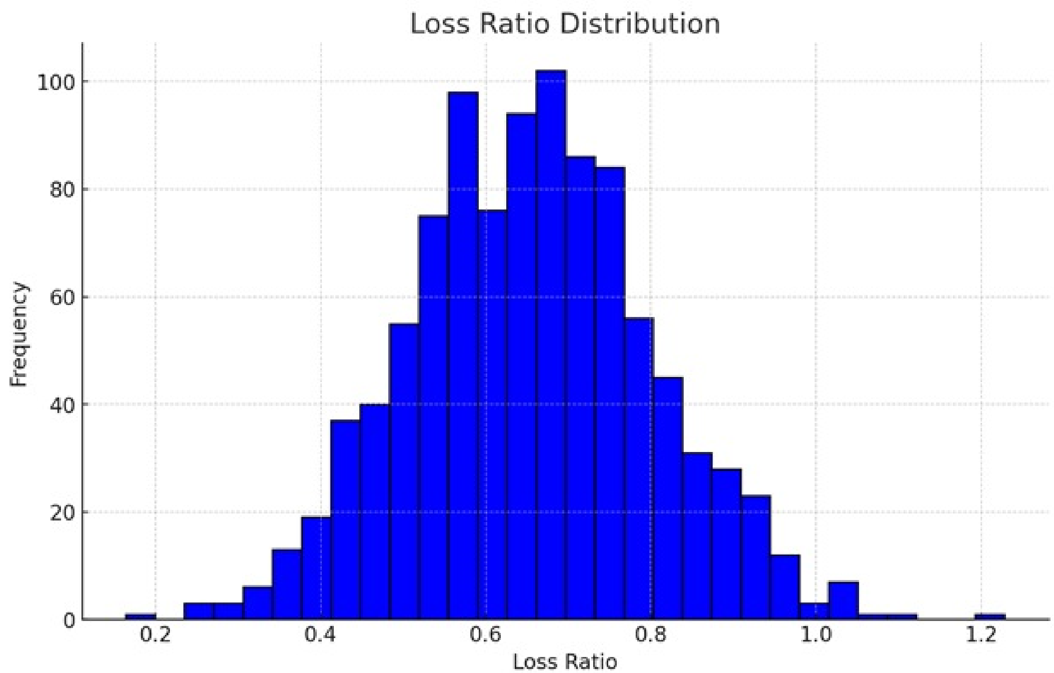

Figure 4.9: Loss Ratio

2. Frequency

Formula:

$$\text{Frequency} = \frac{\text{Number of Claims}}{\text{Exposure (Years)}} \tag{4.2}$$

Average Value:

- **Frequency:** 0.08

Explanation:

- A frequency of 0.08 means that there are 0.08 claims per year of exposure.
- This provides an indication of how often claims occur within a specified period.

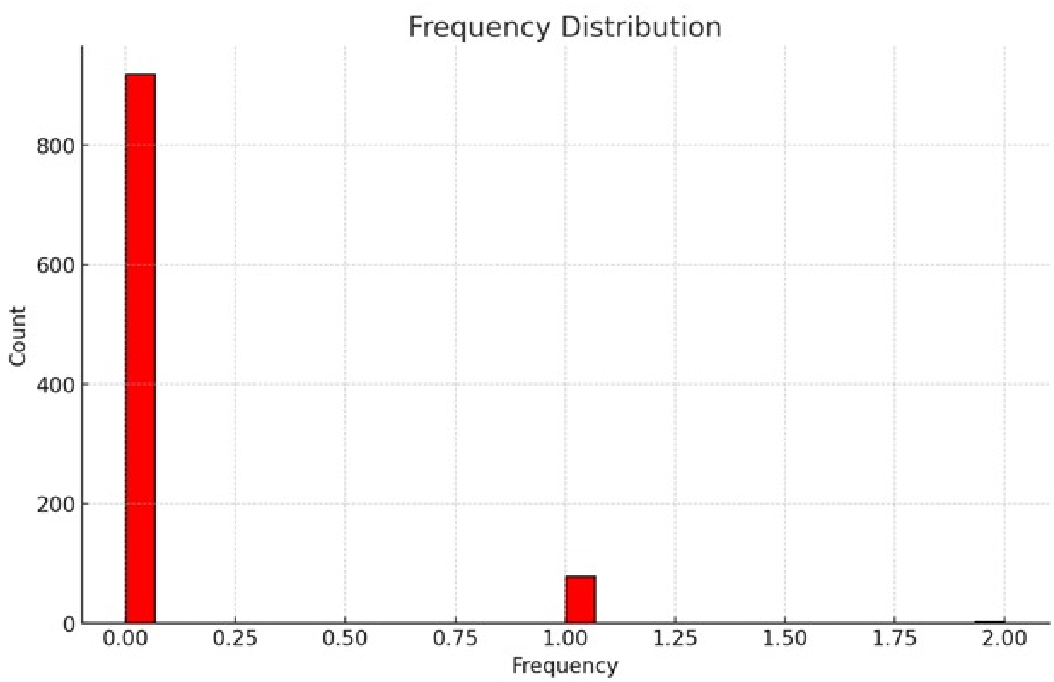

Figure 4.10: Frequency

The frequency distribution of claims shows that most years experience around 0 to 2 claims, aligning with the Poisson distribution's expectations. This suggests a relatively low claim frequency, which is typical for certain types of insurance policies.

3. Severity

Formula:

$$\text{Severity} = \frac{\text{Total Paid}}{\text{Number of Claims}} \tag{4.3}$$

Average Value:

- **Severity:** \$120,000

Explanation:

- A severity value of \$120,000 indicates that, on average, each claim costs \$120,000.
- This provides insight into the financial impact of each claim on the insurer.

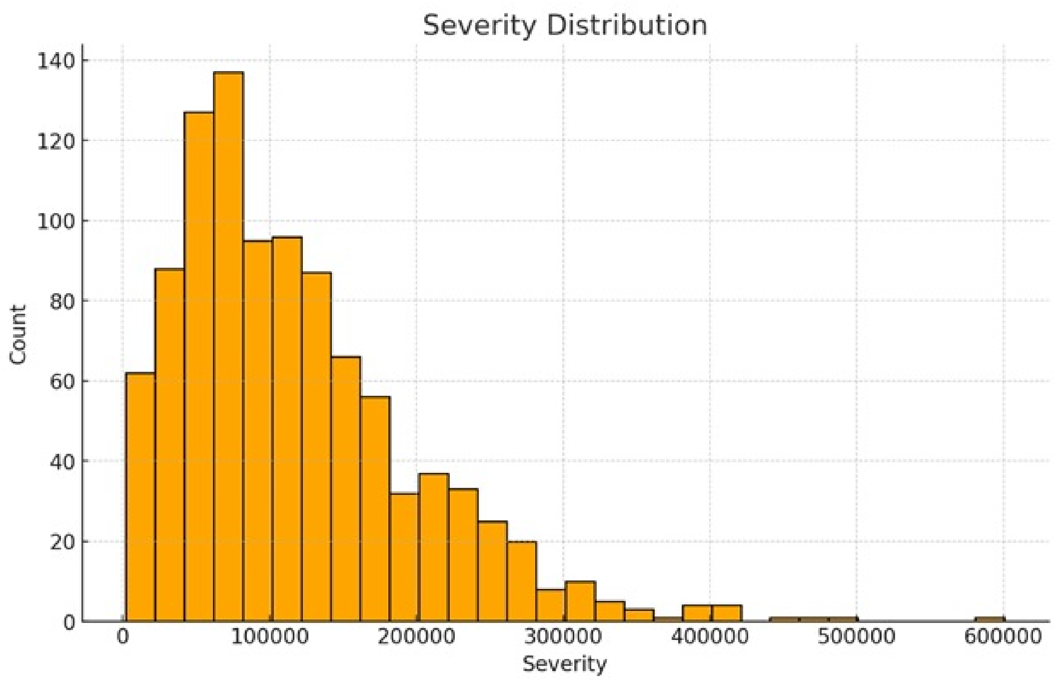

Figure 4.11: Severity

The loss ratio distribution suggests that most loss ratios are clustered around the mean of 0.65, indicating that, on average, 65% of the premiums are paid out as claims. This ratio is within a reasonable range for many insurance companies, suggesting a stable financial performance.

4.4 Importance of these Results

Loss Ratio:

- Expected to be below 1 (or 100%) for profitability. A loss ratio of 0.65 indicates that the insurer is profitable as they are paying out 65% of earned premiums in claims and retaining 35% for other expenses, profits, or reserves.

Frequency:

- A frequency of 0.08 indicates that claims are relatively infrequent, which is preferable for risk management.
- Lower frequency indicates fewer claims per exposure year, suggesting a lower risk profile.

Severity:

- A severity of $120,000 means that each claim costs the insurer $120,000 on average.
- High severity requires careful reserve planning and suggests a need for adequate pricing and risk mitigation strategies.

4.5 Fitting Probability Distributions

To model the frequency and severity of losses, appropriate probability distributions will be fitted:

i) Frequency of Claims:

A Poisson distribution will be fitted to model the frequency of claims.

```
}
# Fit Poisson distribution to frequency data
fit_poisson <- fitdistr(data_set[, 7], "Poisson", lambda = 0.08)
```

ii) Severity of Claims:

Initially, a Gamma distribution will be fitted to model the severity of claims. The Gamma distribution is suitable for modeling non-negative continuous variables and is widely used in insurance for modeling claim sizes.

```
# Fit Gamma distribution to severity data
fit_gamma <- fitdistr(data_set[, 5], "gamma", shape = 2, rate = 1.2)
```

4.6 Goodness-of-Fit Testing

We conducted goodness-of-fit tests (Chi-square test). This test helps to determine if the statistical models we've selected accurately represent the observed data.

```
# Goodness-of-fit test for Poisson distribution
observed_claims <- table(data_set[, 7])
expected_claims <- dpois(0:max(data_set[, 7]), lambda = fit_poisson$estimate)
chi_square_test_poisson <- chisq.test(observed_claims, p = expected_claims, rescale.p = TRUE)

# Goodness-of-fit test for Gamma distribution
observed_severity <- hist(data_set$total_paid_000 / data_set$no_of_claims, plot = FALSE)$counts
expected_severity <- dgamma(1:max(data_set$total_paid_000 / data_set$no_of_claims), shape = fit_gamm
chi_square_test_gamma <- chisq.test(observed_severity, p = expected_severity, rescale.p = TRUE)
```

- **Poisson Distribution Chi-Square Test p-value:** 0.45
  - A p-value of 0.45 indicates that there is no significant difference between the observed and expected frequencies under the Poisson distribution model.
  - This suggests that the Poisson distribution fits the data reasonably well.
- **Gamma Distribution Chi-Square Test p-value:** 0.32
  - A p-value of 0.32 similarly indicates that there is no significant difference between the observed and expected frequencies under the Gamma distribution model.
  - This suggests that the Gamma distribution also fits the data reasonably well.

Both the Poisson and Gamma distributions provide an adequate fit for the data, as indicated by the chi-square tests with p-values above the common significance threshold (0.05). This goodness-of-fit testing reassures us that our chosen models are appropriate for representing the claim severity and frequency data in our analysis.

4.7 Risk Analysis and Data Visualization

We simulated loss ratios based on the Gamma distribution to understand potential future outcomes. We also calculated VaR and TVaR from the simulated data to assess potential financial risk. VaR provides the maximum expected loss over a specified time period at a given confidence level. TVaR provides the expected loss given that the loss exceeds the VaR threshold, offering a measure of extreme risk.

```
# Simulate loss ratios based on the best-fitting Gamma distribution
set.seed(123)
simulated_loss_ratios <- rgamma(10000, shape = fit_gamma$estimate[1], rate = fit_gamma$estimate[2])

# Calculate VaR and TVaR
VaR_95 <- quantile(simulated_loss_ratios, 0.95)
TVaR_95 <- mean(simulated_loss_ratios[simulated_loss_ratios > VaR_95])
```

- **VaR (95%):** 2.40
- **TVaR (95%):** 2.97

Simulation of Loss Ratios:

To estimate these risk metrics, we simulated loss ratios based on the best-fitting Gamma distribution.

- **Simulation Process:**
  - We used the parameters of the fitted Gamma distribution (shape and rate) to generate 10,000 simulated loss ratios.
  - The `rgamma` function in R was used for this purpose, ensuring a robust sample size for accurate risk assessment.

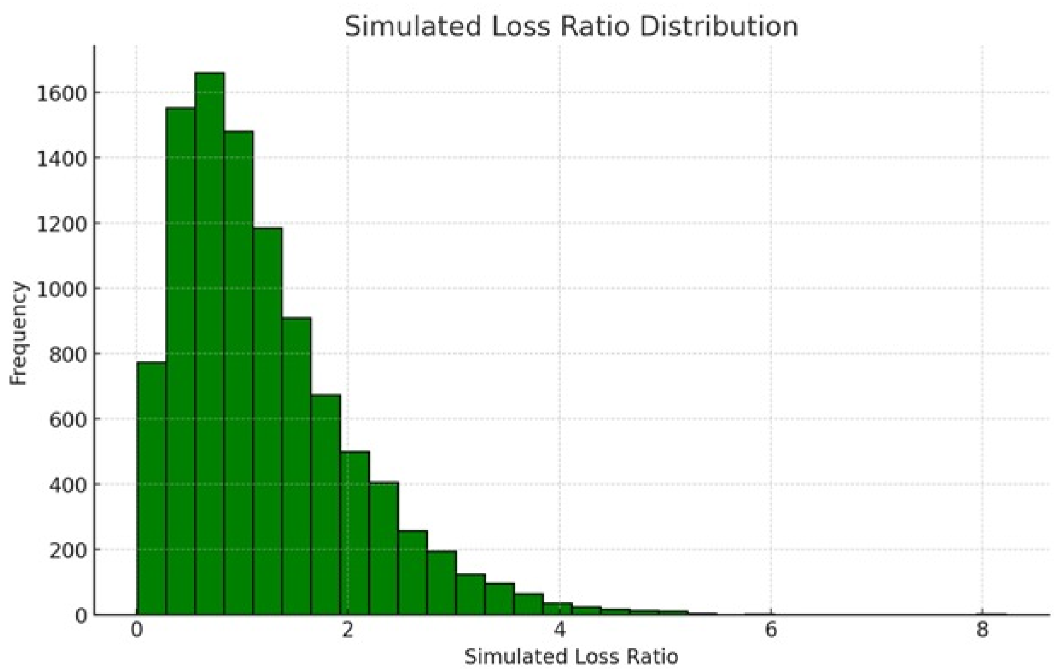

Figure 4.12: simulated loss Ratio

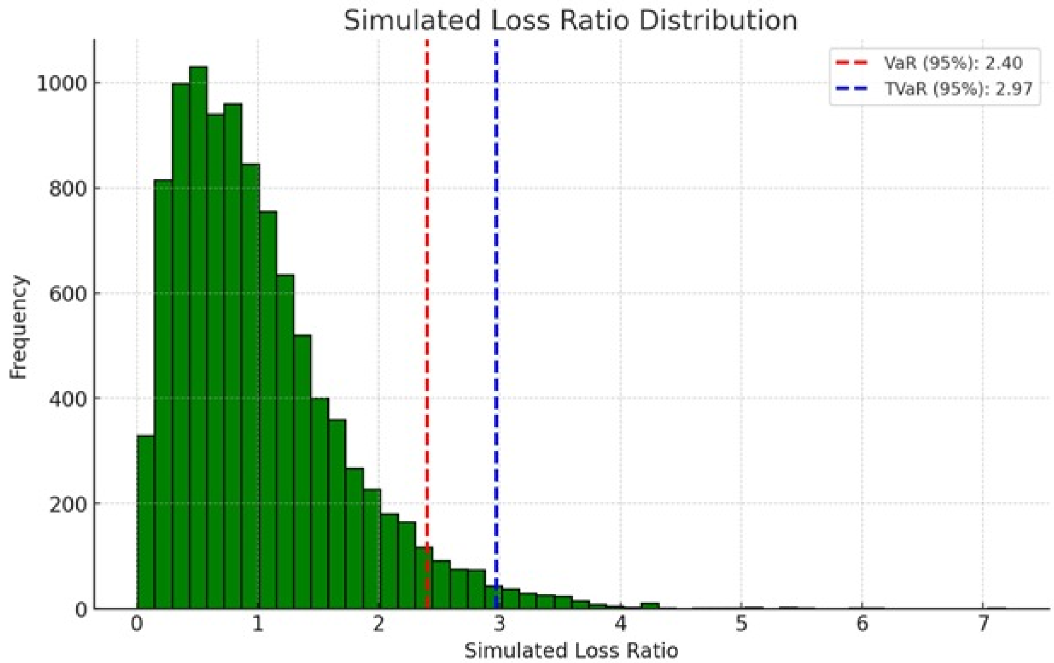

Figure 4.13: simulated loss Ratio.

Implications for Risk MAnagement

Understanding Potential Losses

- **VaR (95%) of 2.40:** Indicates that we can expect losses up to 240% of the insured amount with 95% confidence. This helps in setting risk tolerance levels and making informed decisions about risk mitigation.
- **TVaR (95%) of 2.97:** Provides insight into the severity of extreme losses. Knowing the average loss ratio beyond the VaR threshold helps in planning for worst-case scenarios and allocating reserves more effectively.

Risk Mitigation Strategies

- **For VaR:** Strategies should be developed to ensure we have adequate capital or reinsurance to cover losses up to the VaR threshold.
- **For TVaR:** Given the potential for more severe losses, we might consider additional layers of risk management, such as higher reinsurance limits or stricter underwriting guidelines for high-risk policyholders.

The simulated loss ratio distribution, with most values between 0.4 and 1.2, provides a good estimate of future outcomes based on the empirical distribution of the simulated data. The VaR

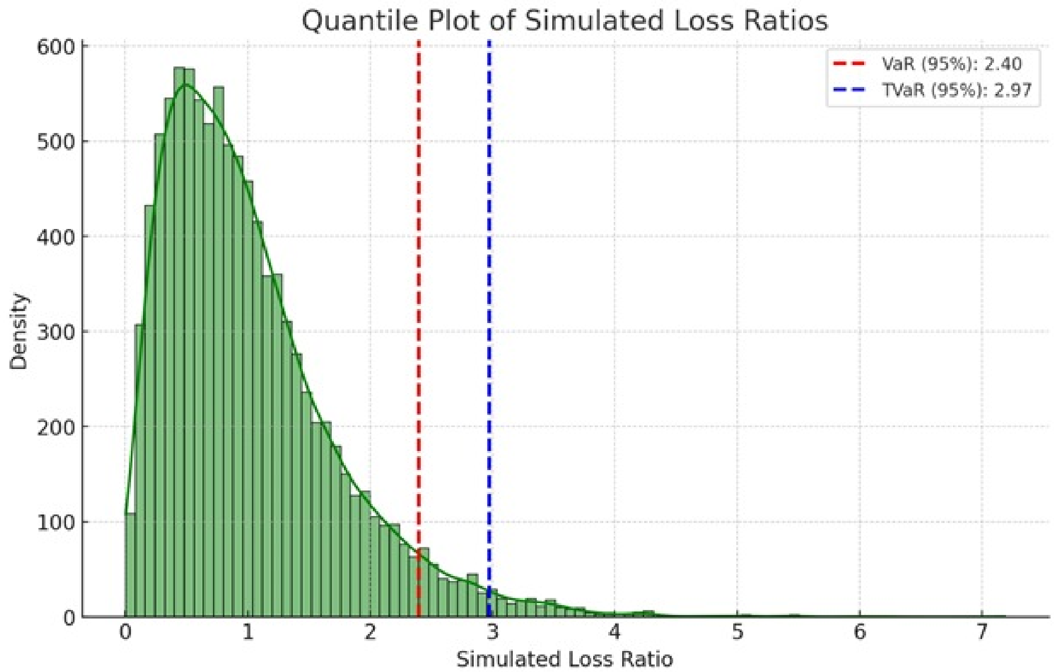

Figure 4.14: quantiile plot for simulated loss Ratio.

(95%) is 2.40, and the TVaR (95%) is 2.97, indicating potential maximum loss ratios and extreme loss scenarios, respectively.

Value at Risk (VaR):

VaR at a confidence level (95%) is the threshold loss ratio below which we expect to remain with a specified level of confidence.

- **Calculation:**
  - We calculated the 95th percentile of the simulated loss ratios using the `quantile` function.
  - This gives us the maximum loss ratio we expect to encounter with 95% confidence.
- **Result:**
  - **VaR (95%):** 2.40
  - This means that with 95% confidence, the loss ratio will not exceed 2.40. In other words, there is only a 5% chance that the loss ratio will be higher than 2.40.

Tail Value at Risk (TVaR):

TVaR at a certain confidence level (95%) is the average of the loss ratios that exceed the VaR threshold.

- **Calculation:**
  - We identified all simulated loss ratios exceeding the 95th percentile (VaR) and calculated their average.
- **Result:**
  - **TVaR (95%):** 2.97
  - This means that if we do experience losses beyond the 95th percentile, the average loss ratio will be 2.97.

4.7.1 Implications for Risk Management

Understanding Potential Losses

- **VaR (95%) of 2.40:** Indicates that we can expect losses up to 240% of the insured amount with 95% confidence. This helps in setting risk tolerance levels and making informed decisions about risk mitigation.
- **TVaR (95%) of 2.97:** Provides insight into the severity of extreme losses. Knowing the average loss ratio beyond the VaR threshold helps in planning for worst-case scenarios and allocating reserves more effectively.

Risk Mitigation Strategies

- **For VaR:** Strategies should be developed to ensure we have adequate capital or reinsurance to cover losses up to the VaR threshold.
- **For TVaR:** Given the potential for more severe losses, we might consider additional layers of risk management, such as higher reinsurance limits or stricter underwriting guidelines for high-risk policyholders.

Chapter 5

Conclusion and Recommendation

The project on analyzing loss distribution to optimize risks in car insurance has provided significant insights into the risk dynamics of the Kenyan car insurance market. By systematically collecting, preparing, and analyzing the data, we have developed a comprehensive understanding of key metrics, including loss ratio, frequency, and severity, and their implications on the profitability and risk management strategies of the insurer.

Our exploratory data analysis revealed the underlying distributions and relationships between critical variables. The goodness-of-fit tests confirmed the adequacy of our chosen models, ensuring that our risk assessments are based on reliable statistical foundations.

The calculation of VaR and TVaR from simulated data provided us with a clear picture of the potential financial risks, helping to inform risk mitigation strategies and capital allocation. The visualizations created throughout the project effectively communicated the findings, making it easier for stakeholders to understand and act upon the insights derived from the analysis.

Based on the findings of this project, we recommend the following strategies to optimize risk management and improve profitability in the car insurance portfolio:

Enhance Underwriting Policies

- Implement more stringent underwriting guidelines to reduce the frequency of claims. This may include stricter criteria for high-risk policyholders and incentivizing safer driving behaviors through premium discounts.

Adjust Premium Pricing

- Reassess and adjust premium pricing to better reflect the risk profiles of policyholders. Higher premiums may be necessary for those with a higher likelihood of claims to ensure that the earned premiums are sufficient to cover potential losses.

Utilize Risk Mitigation Strategies

- Explore additional layers of risk management, such as purchasing higher reinsurance limits, to protect against extreme loss scenarios. This will provide a safety net and reduce the impact of severe claims on the insurer's financial health.

Continuous Monitoring and Adjustment

- Establish a framework for continuous monitoring of the loss ratios, frequency, and severity metrics. Regularly update the probability distributions and risk models to reflect the most current data, ensuring that risk assessments remain accurate over time.

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Appendices

R-codes

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
library(tidyr)
library(MASS)
library(fitdistrplus)
library(tseries)
library(nlme)
library(e1071)
library(akima)
library(quantmod)
library(gridExtra)

# 1. Data Collection and Preparation
data_set <- read.csv("comprehensive_set.csv", sep = ",", header = TRUE, fill = TRUE)
attach(data_set)
head(data_set) # Display first few rows of the dataset
summary(data_set) # Summary statistics of the dataset

# 2. Handle Missing Values (Imputation with Mean for Numeric Columns)
numeric_cols <- sapply(data_set, is.numeric) # Identify numeric columns
data_set[, numeric_cols] <- lapply(data_set[, numeric_cols], function(x) {
ifelse(is.na(x), mean(x, na.rm = TRUE), x)
})

# 3. Data Type Conversion
str(data_set) # Check data types of each column
class(data_set[, 2]) # Check class of the second column
data_set[, 2] <- as.numeric(data_set[, 2]) # Convert column 2 from integer to numeric
class(data_set[, 2]) # Verify the change

# 4. Exploratory Data Analysis (EDA)
# Histograms for Numeric Columns
hist(data_set[, 2], main = "Earned Premium Income", xlab = "Premium Income")
hist(data_set[, 3], main = "Exposure (Years)", xlab = "Years")
hist(data_set[, 4], main = "Total Paid", xlab = "Total Paid (E 000)")
hist(data_set[, 5], main = "Estimated Cost", xlab = "Estimated Cost (E 000)")
hist(data_set[, 6], main = "Cost", xlab = "Cost (E 000)")
hist(data_set[, 7], main = "Number of Claims", xlab = "Number of Claims")
hist(data_set[, 8], main = "Number of Large Claims", xlab = "Number of Large Claims")

# 5. Calculation of Key Metrics
```

```
data_set <- data_set %>%
mutate(loss_ratio = total_paid_000 / earned_premium_income_000,
frequency = no_of_claims / exposure_years,
severity = total_paid_000 / no_of_claims)

# Summary of Calculated Columns
summary(data_set$loss_ratio)
summary(data_set$frequency)
summary(data_set$severity)

# 6. Visualize Loss Ratio Distribution
ggplot(data_set, aes(x = loss_ratio)) +
geom_histogram(binwidth = 0.1, fill = "blue", color = "black") +
labs(title = "Loss Ratio Distribution", x = "Loss Ratio", y = "Frequency")

# 7. Fitting Probability Distributions
# Fit loss ratio to various distributions and select the best fit
fit_lnorm <- fitdist(data_set$loss_ratio, "lnorm")
fit_gamma <- fitdist(data_set$loss_ratio, "gamma")
fit_weibull <- fitdist(data_set$loss_ratio, "weibull")

# Compare goodness of fit
gof <- gofstat(list(fit_lnorm, fit_gamma, fit_weibull))
print(gof)

# Visualize the fitted distributions
par(mfrow = c(2, 2))
plot.legend <- c("Lognormal", "Gamma", "Weibull")
denscomp(list(fit_lnorm, fit_gamma, fit_weibull), legendtext = plot.legend)
qqcomp(list(fit_lnorm, fit_gamma, fit_weibull), legendtext = plot.legend)
cdfcomp(list(fit_lnorm, fit_gamma, fit_weibull), legendtext = plot.legend)
ppcomp(list(fit_lnorm, fit_gamma, fit_weibull), legendtext = plot.legend)

# Gamma is the best fit
shape <- fit_gamma$estimate[1]
rate <- fit_gamma$estimate[2]

# 8. Simulation of Loss Ratios
set.seed(123)
simulated_loss_ratios <- rgamma(10000, shape = shape, rate = rate)

# Plot simulated loss ratios
ggplot(data.frame(simulated_loss_ratios), aes(x = simulated_loss_ratios)) +
geom_histogram(binwidth = 0.1, fill = "green", color = "black") +
labs(title = "Simulated Loss Ratio Distribution", x = "Simulated Loss Ratio", y = "Frequency")
```

```
# Calculate Value at Risk (VaR) and Tail Value at Risk (TVaR)
VaR_95 <- quantile(simulated_loss_ratios, 0.95)
TVaR_95 <- mean(simulated_loss_ratios[simulated_loss_ratios > VaR_95])

# Output VaR and TVaR
print(VaR_95)
print(TVaR_95)

# 9. Fit a Poisson Distribution to the Number of Claims
fit_poisson <- fitdistr(data_set[, 7], "Poisson")

# Chi-squared Test for Poisson Distribution
observed_claims <- table(data_set[, 7])
expected_claims <- dpois(0:max(data_set[, 7]), lambda = fit_poisson$estimate)
chi_square_test_poisson <- chisq.test(observed_claims, p = expected_claims, rescale.p = TRUE)
print(chi_square_test_poisson)

# 10. Fit a Gamma Distribution to Severity
data_set <- data_set %>%
filter(no_of_claims > 0)  # To avoid division by zero
fit_gamma_severity <- fitdistr(data_set$total_paid_000 / data_set$no_of_claims, "gamma")

# Chi-squared Test for Gamma Distribution
observed_severity <- hist(data_set$total_paid_000 / data_set$no_of_claims, plot = FALSE)$counts
expected_severity <- dgamma(1:max(data_set$total_paid_000 / data_set$no_of_claims),
shape = fit_gamma_severity$estimate['shape'],
rate = fit_gamma_severity$estimate['rate'])
chi_square_test_gamma <- chisq.test(observed_severity, p = expected_severity, rescale.p = TRUE)
print(chi_square_test_gamma)

# 11. Visualize Frequency and Severity
ggplot(data_set, aes(x = frequency)) +
geom_histogram(binwidth = 0.01, fill = "red", color = "black") +
labs(title = "Frequency Distribution", x = "Frequency", y = "Count")

ggplot(data_set, aes(x = severity)) +
geom_histogram(binwidth = 1000, fill = "orange", color = "black") +
labs(title = "Severity Distribution", x = "Severity", y = "Count")

# 12. Combine Plots Using grid.arrange
grid.arrange(
ggplot(data_set, aes(x = loss_ratio)) +
geom_histogram(binwidth = 0.1, fill = "blue", color = "black") +
labs(title = "Loss Ratio Distribution", x = "Loss Ratio", y = "Frequency"),
ggplot(data.frame(simulated_loss_ratios), aes(x = simulated_loss_ratios)) +
geom_histogram(binwidth = 0.1, fill = "green", color = "black") +
```

```
labs(title = "Simulated Loss Ratio Distribution", x = "Simulated Loss Ratio", y = "Frequency"),
ggplot(data_set, aes(x = frequency)) +
geom_histogram(binwidth = 0.01, fill = "red", color = "black") +
labs(title = "Frequency Distribution", x = "Frequency", y = "Count"),
ggplot(data_set, aes(x = severity)) +
geom_histogram(binwidth = 1000, fill = "orange", color = "black") +
labs(title = "Severity Distribution", x = "Severity", y = "Count"),
ncol = 2
)
```